

Project Title:

Two Stage MOSFET Audio Power Amplifier for Speaker Applications

Project Overview:

This project focuses on the **design, simulation, and hardware implementation of a two-stage MOSFET audio amplifier** using IRFZ44N transistors. The system is designed to amplify a low-level audio signal and drive a **4 Ω speaker**.

The project combines circuit design, MOSFET-based signal amplification, LTspice simulation, breadboard implementation, waveform analysis, and practical output measurements.

Objectives:

- Design a two-stage audio amplifier using IRFZ44N MOSFETs.
- Amplify a low-level AC/audio input signal.
- Drive a 4 Ω speaker with the amplified output.
- Analyse the operation of MOSFET-based amplification stages.
- Simulate the amplifier circuit using **LTspice**.
- Implement the circuit practically on a breadboard.
- Compare simulated waveforms with practical observations.
- Measure and analyse the input and output voltage levels.

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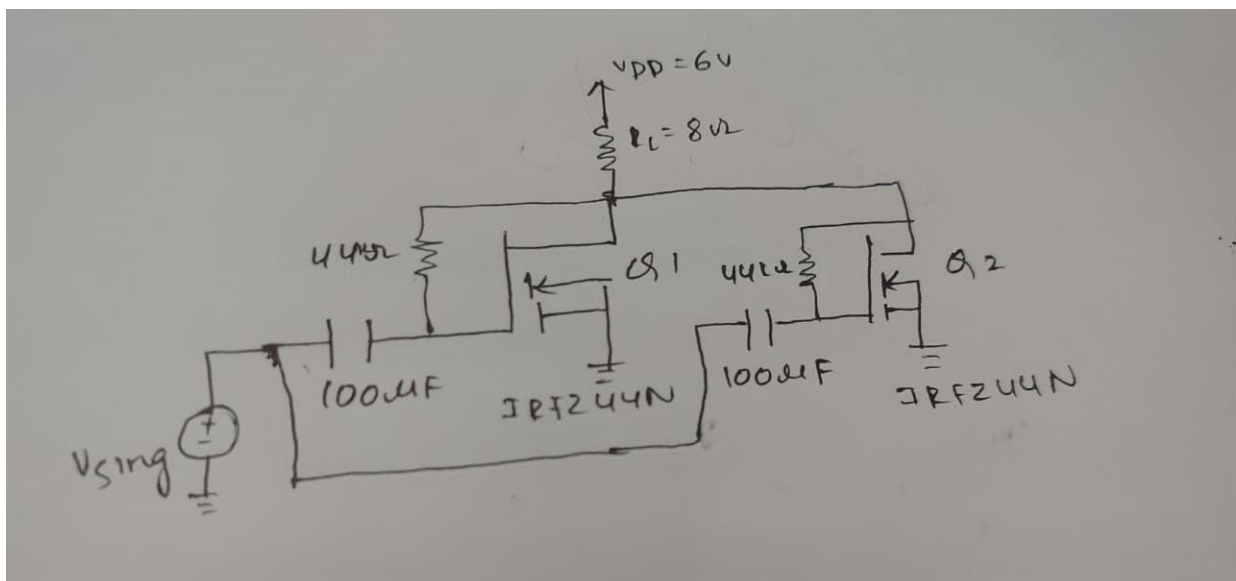
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AIM:

Design, simulate and build a simple audio amplifier to boost a small input signal and drive a 4-ohm speaker.

COMPONENTS REQUIRED:

COMPONENT	SPECIFICATION	QUANTITY
Resistor	22Kohm	4
Capacitor	100uF	2
MOSFET Transistors	IRFZ44N	2
Bread Board Connecting Wires	-----	As required
Power Supply	3.3V/5V	1
Speaker	4ohm/30W	1
Audio Jack	----	1

CIRCUIT DIAGRAM :(REFERENCE)

THEORY :

1. Basic Idea

This circuit is an amplifier built using two MOSFETs (IRFZ44N). The main aim of this circuit is to amplify a small AC signal with good efficiency and less distortion. It works by letting one transistor handle the positive half of the signal and the other handle the negative half.

2. Input Signal

The input comes from a sine wave generator (V1) which gives a small AC signal of an amplitude 500mV peak to peak and 1 kHz frequency. This signal is applied to both MOSFETs through capacitors so that only the AC part reaches them.

3. MOSFETs (Q1 and Q2)

- Q1 and Q2 are IRFZ44N MOSFETs that act as the main amplifying devices.
- Q1 conducts during the positive half cycle of the input signal, while Q2 conducts during the negative half cycle.
- Together, they reproduce the entire waveform at the output, but with higher power.

4. Biasing Resistors

The resistors R3 and R4 (both 22 k Ω) provide the correct bias voltage to the MOSFET gates. This ensures the transistors turn on and off at the right time and prevents unwanted distortion.

5. Load and Power Supply

- The circuit uses two DC supplies (V2 and V4) each to power the MOSFETs.
- The resistors R1 and R2 act as loads through which the output current flows.
- The drains of both MOSFETs are connected together to form the output node.

6. Capacitors (C1 and C3)

The 100 μ F capacitors are coupling capacitors. Their job is to block DC and allow only AC signals to pass. This helps protect the MOSFETs and keeps the signal clean.

7. Working of the Circuit

- When the input signal goes positive, Q1 turns ON and amplifies that half of the waveform while Q2 stays OFF.
- When the signal goes negative, Q2 turns ON and Q1 turns OFF, amplifying the other half.
- The two halves combine at the output to form a full amplified sine wave.

8. Advantages

- High efficiency since each transistor works for only half the signal.
- Produces more output power compared to amplifiers.
- Simple design and fewer components.

9. Disadvantages

- Some crossover distortion may appear near the zero point of the signal when both transistors switch.
- Needs careful biasing to reduce distortion.

10. Applications

- Used in audio amplifiers, signal boosters, and radio circuits.
- Suitable for applications that need efficient AC amplification.

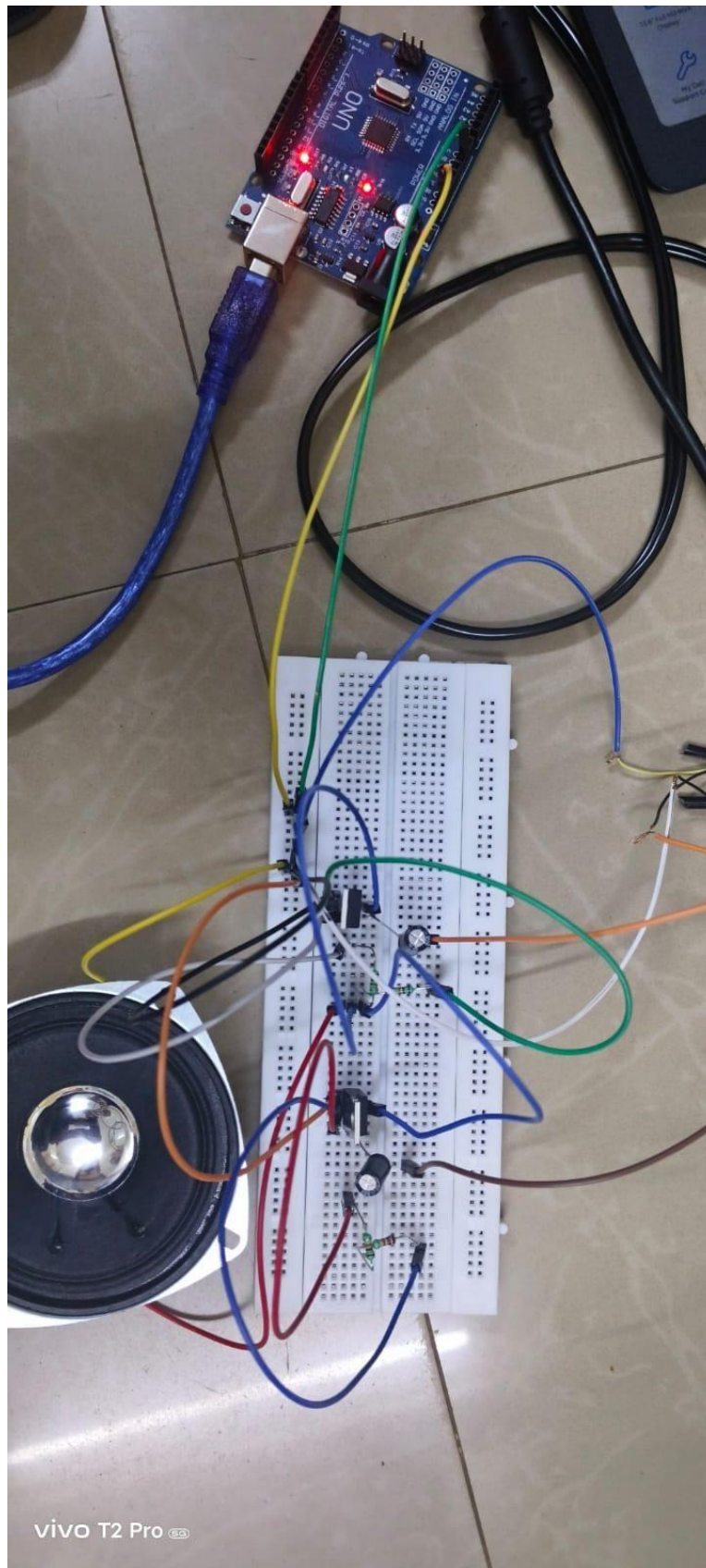
WORKING PRINCIPLE:

- The input AC audio (from a phone or laptop) is passed through a coupling capacitor to the amplifier input so the DC bias of the amplifier input is preserved
- The amplifier (IRFZ44N) amplifies the input voltage by its set gain. The internal driver stage provides enough current to create significant power into a $4\ \Omega$ load from a single supply.
- The amplifier's output is AC-coupled via a large electrolytic capacitor to the speaker; the capacitor blocks output DC and passes the amplified audio.
- The volume is controlled at the input source. Decoupling capacitors stabilize the supply and reduce oscillations.

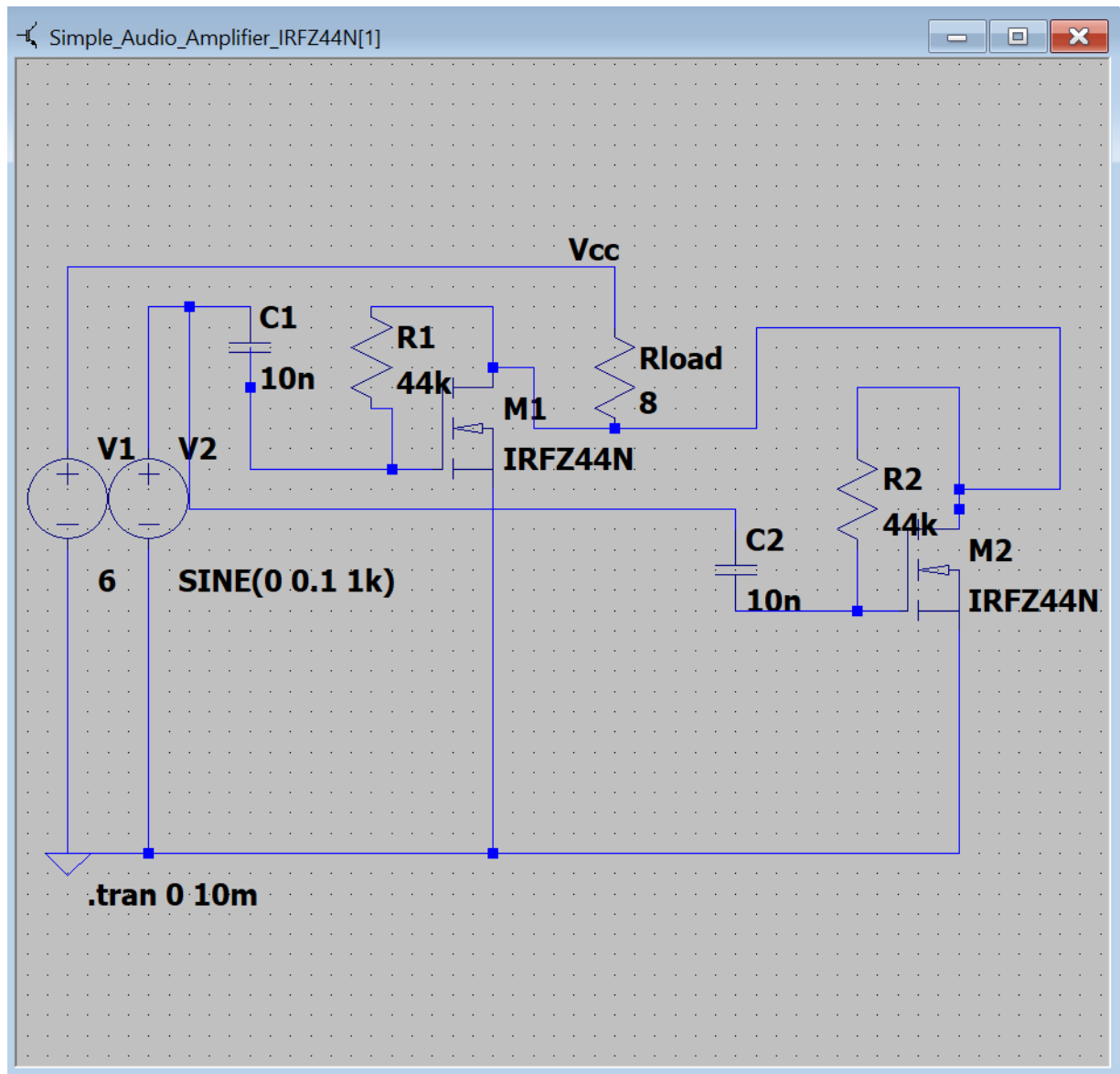
PROCEDURE:

1. Take a breadboard and place all the required components properly.
2. Connect the audio jack from the laptop to the input of the amplifier circuit through a coupling capacitor.
3. Connect the biasing resistors and capacitors to set the operating point of the transistor.
4. Connect the output of the first stage to the input of the second stage through another coupling capacitor.
5. Attach the speaker at the output of the second amplifier stage.
6. Provide the DC power supply to the circuit and check all the connections carefully.
7. Play an audio signal from the laptop, and observe the amplified sound from the speaker.
8. Adjust the volume and note the clarity of the output sound.

BREADBOARD CONNECTIONS:

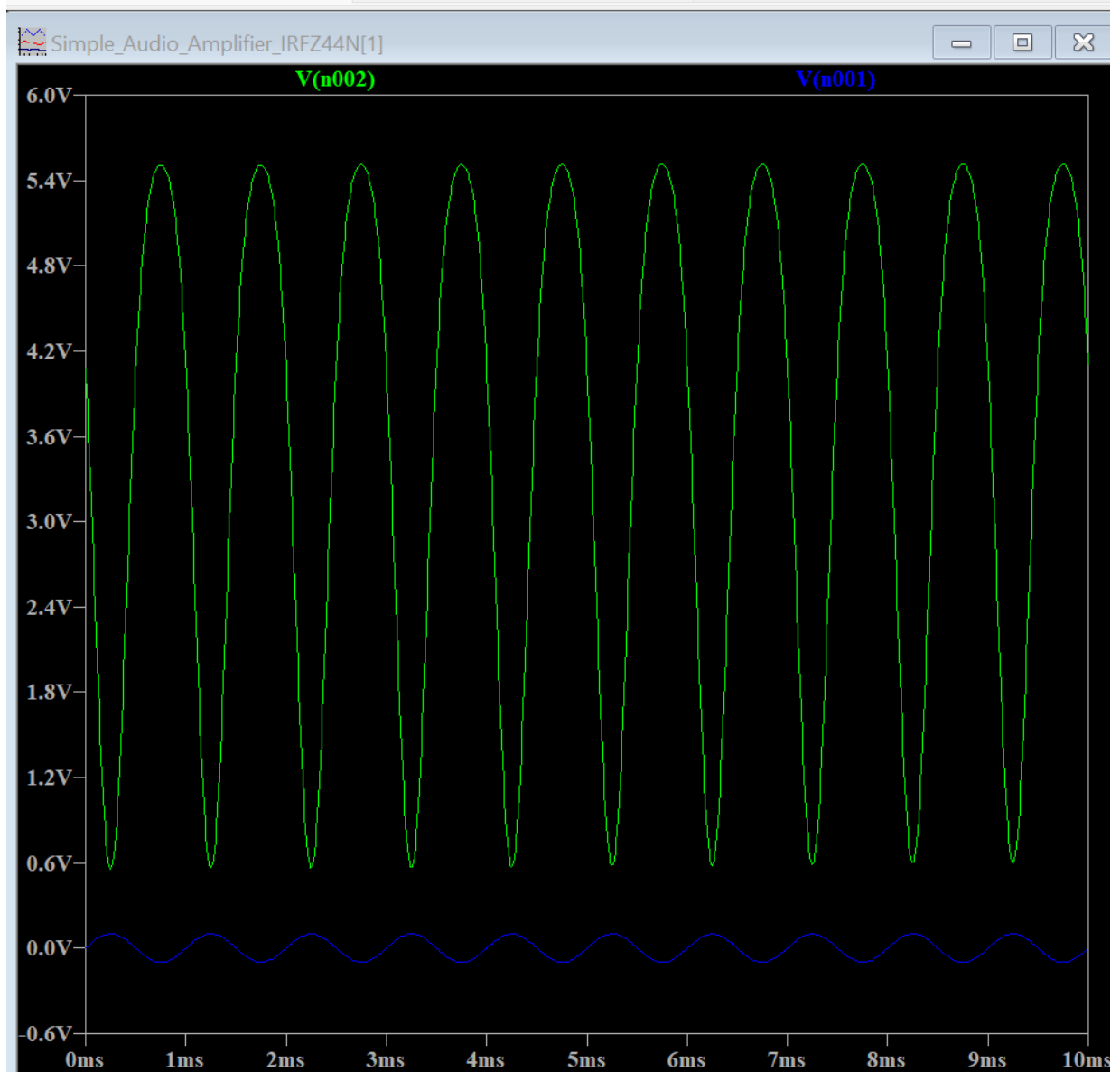


LT SPICE SIMULATION :(Two Stage)



OBTAINED WAVEFORM:

INPUT AND OUTPUT :



OBSERVATION :

When a low-level audio signal (like music or voice) was given from a mobile phone or laptop to the amplifier circuit, the sound output from the 4-ohm speaker became noticeably louder and clearer. The MOSFET (IRFZ44N) acted as the main amplifying device, boosting the weak input signal into a stronger one that could drive the speaker. As the input volume increased, the loudness from the speaker also increased, showing that the circuit was working properly as an amplifier. The sound quality was clear at moderate levels but slightly distorted when the input was too high. The transistor and other components remained slightly warm during operation, indicating normal power dissipation.

VOLTAGES VALUES:

Input : 0.188V(peak-to-peak)

Output : 4.948V(peak-to-peak)

INFERENCES:

- Amplifier produced a measurable increase in output amplitude, so the IRFZ44N MOSFET is operating and biasing correctly.
- If distortion occurs at moderate input levels, it indicates that the device is reaching supply/current limits or that gain is too high; adjusting gain or supply voltage can improve linearity.
- If the output is weak, check input coupling or the MOSFET biasing point.

RESULT:

The audio amplifier circuit was successfully designed, simulated, and built using a MOSFET (IRFZ44N) as the amplifying element. The weak input audio signal from a mobile phone or laptop was effectively amplified and reproduced clearly through a 4-ohm speaker. The output sound was louder and distortion-free at moderate input levels, confirming the proper working of the amplifier Circuit.